

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a **list of n numbers**. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A is **bigger than** B then this step's **output is 1** . B **removes** the number that was picked from their list.
2. If the number picked by A is **smaller than** B then this step's **output is 2** . A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is 0** . **Both A and B remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

$$1 \leq N \leq 10^6$$

$$1 \leq \text{num} \leq 10^9$$

Input format:

First line consists of a number n , size of the list provided.

Next line consists of n numbers separated by space.

Output format:

Output the required output list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 2 3

EXPECTED OUTPUT

2 2 0

Test Case 2

INPUT (STDIN)

5
5 7 1 2 3

EXPECTED OUTPUT

1 1 1 2 0 2 2 2

Mandatory Test Cases

Test Case 1

KEYWORD

if(a[i]>a[j])

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

160

Test Case 3

NLOC

36

```
Stack1.c +
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6     long long a[n], b[n];
7     for (int k = 0; k < n; k++) {
8         scanf("%lld", &a[k]);
9         b[k] = a[k];
10    }
11
12    int i = 0, j = n - 1;
13    while (i < n && j >= 0) {
14        if (a[i] > a[j]) {
15            printf("1 ");
16            j--;
17        } else if (a[i] < a[j]) {
18            printf("2 ");
19            i++;
20        } else {
21            printf("0 ");
22            i++;
23            j--;
24        }
25    }
```

main | CH.SC.U4CSE25225 | Stack1.c

Ln 29, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     printf("\n");  
27     return 0;
```

28}

29

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

1 2 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack1.exe

2 2 0

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

5

5 7 1 2 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack1.exe

1 1 1 2 0 2 2 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>